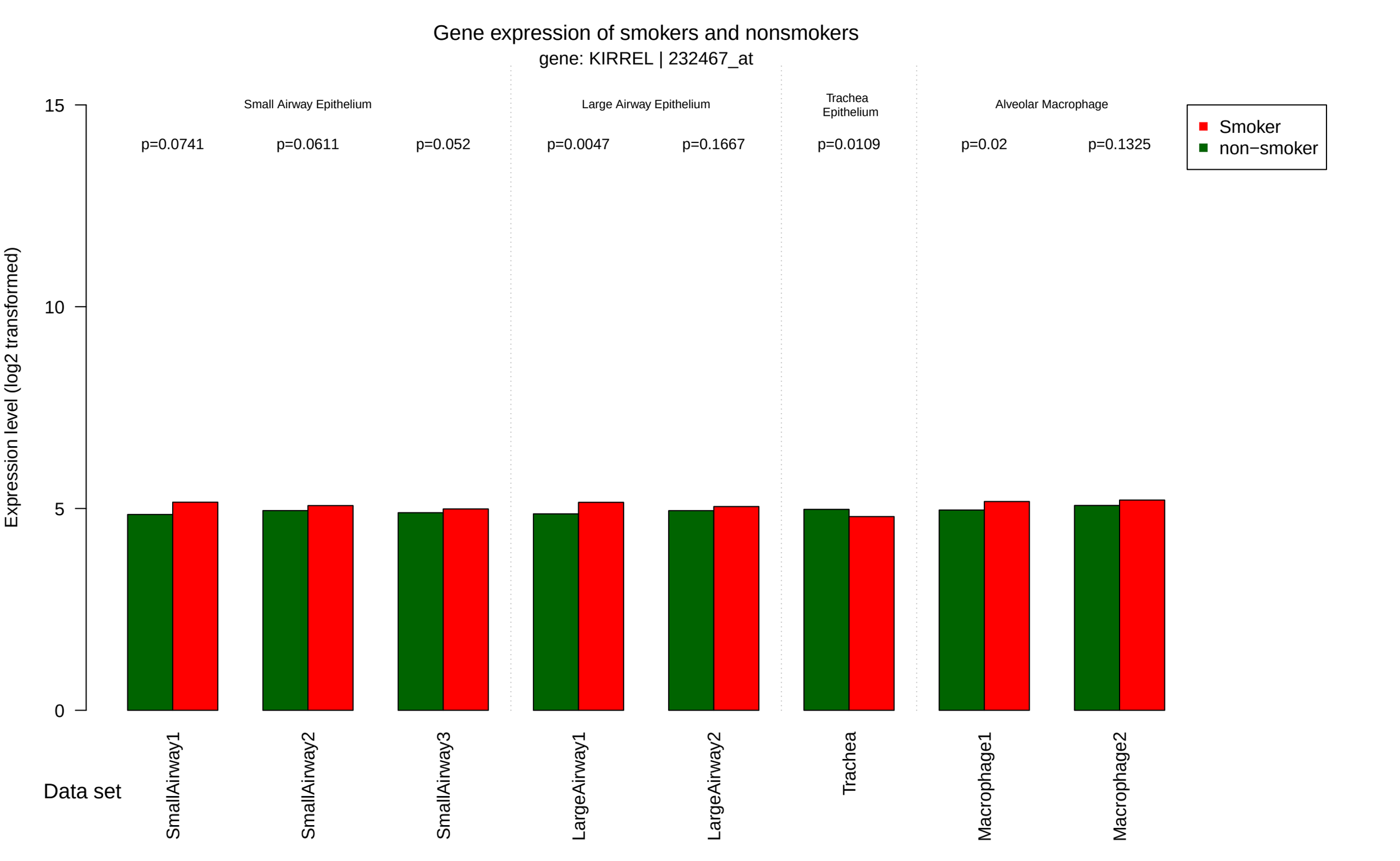
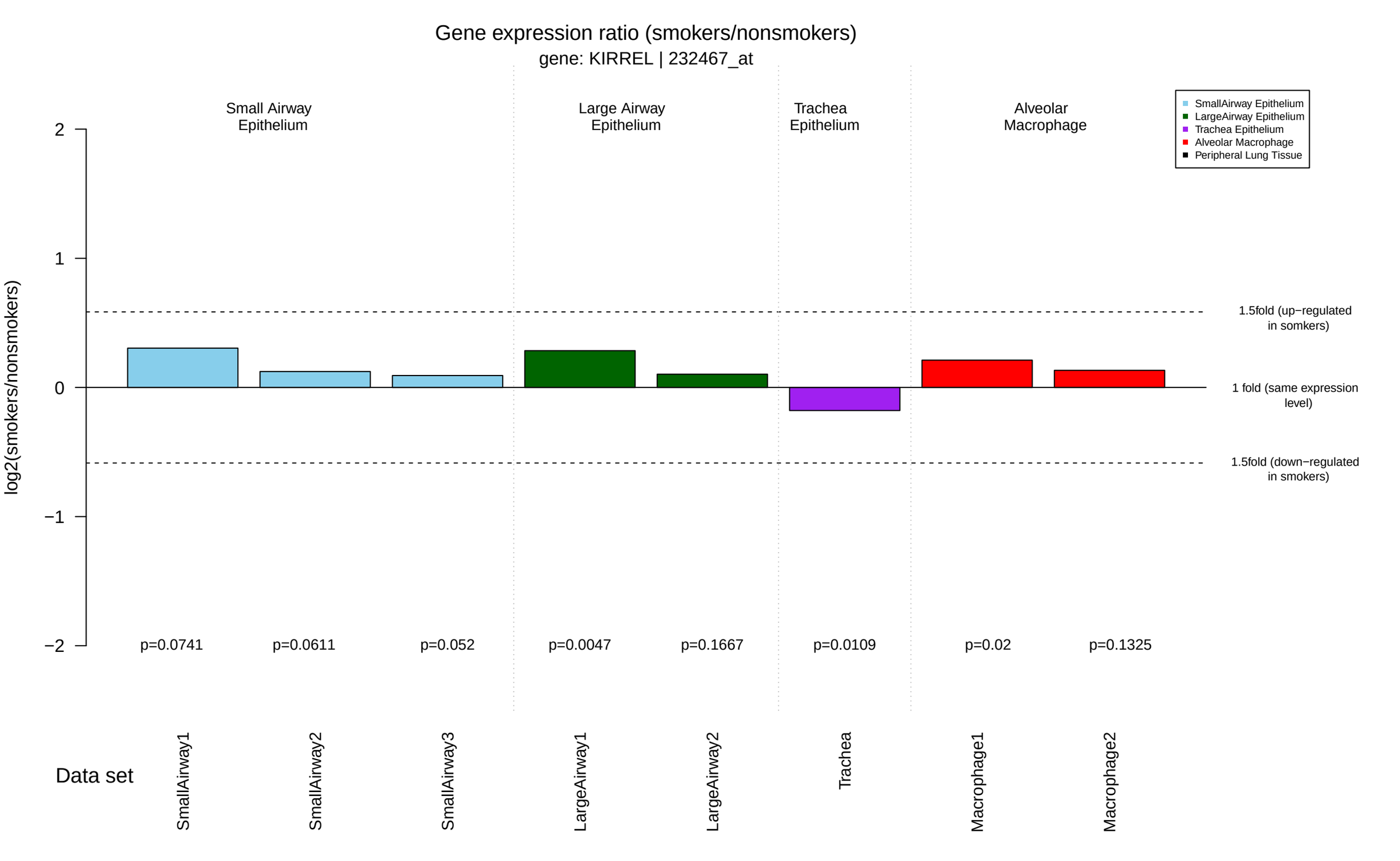
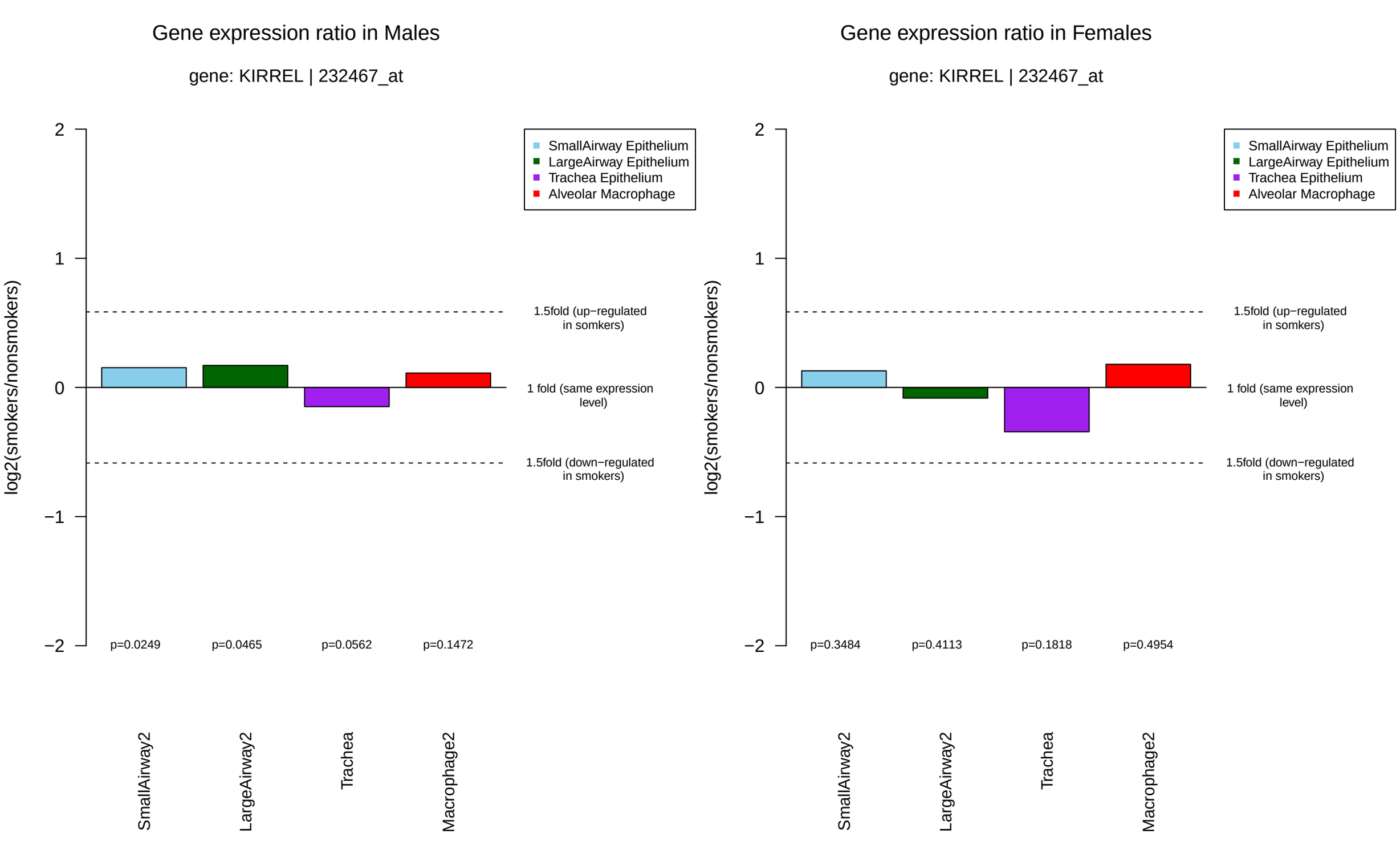
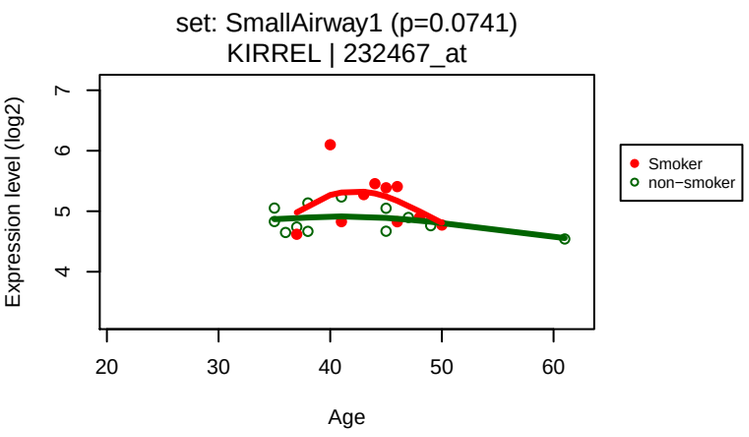
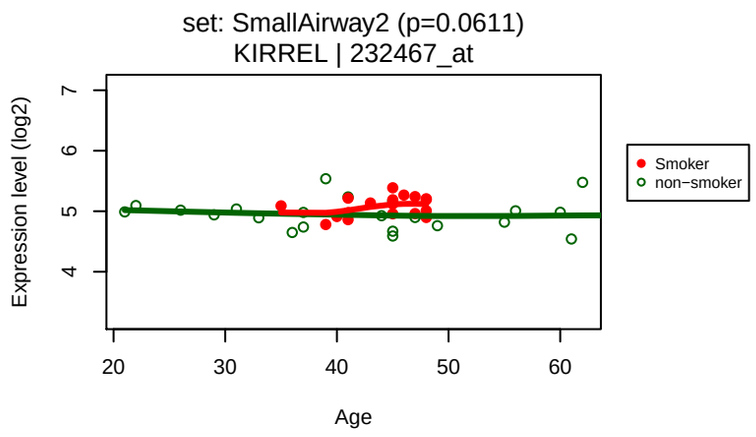
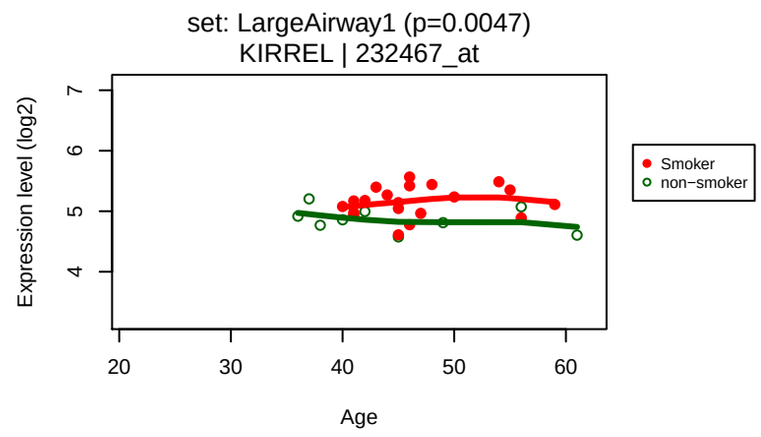

Gene expression of smokers and nonsmokers
gene: KIRREL | 232467_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0741
p=0.0611
p=0.052
p=0.0047
p=0.1667
p=0.0109
p=0.02
p=0.1325
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

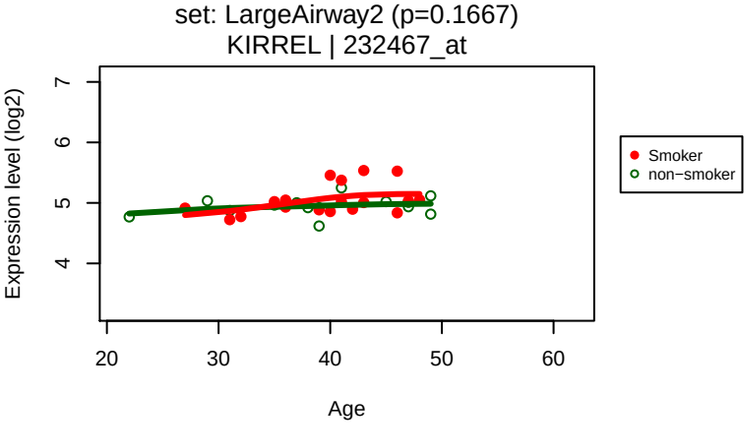

Gene expression ratio (smokers/nonsmokers)
gene: KIRREL | 232467_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0741
p=0.0611
p=0.052
p=0.0047
p=0.1667
p=0.0109
p=0.02
p=0.1325
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

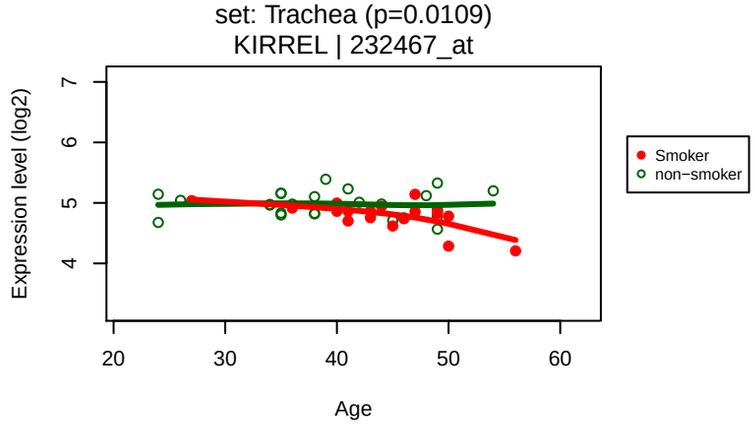

Gene expression ratio in Males
gene: KIRREL | 232467_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0249
p=0.0465
p=0.0562
p=0.1472
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KIRREL | 232467_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3484
p=0.4113
p=0.1818
p=0.4954
SmallAirway2
LargeAirway2
Trachea
Macrophage2

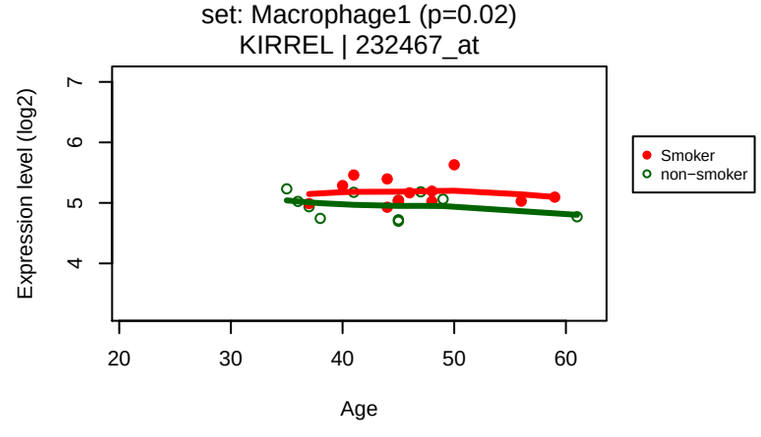

set: SmallAirway1 (p=0.0741)
KIRREL | 232467_at
Expression level (log2)
Age
Smoker
non−smoker

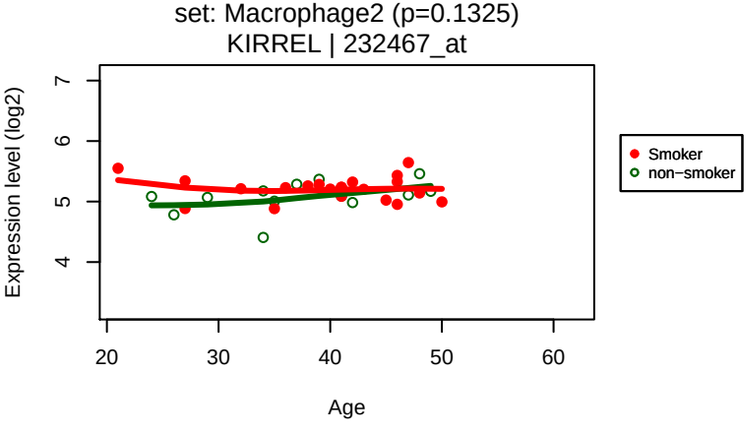

set: SmallAirway2 (p=0.0611)
KIRREL | 232467_at
Expression level (log2)
Age
Smoker
non−smoker

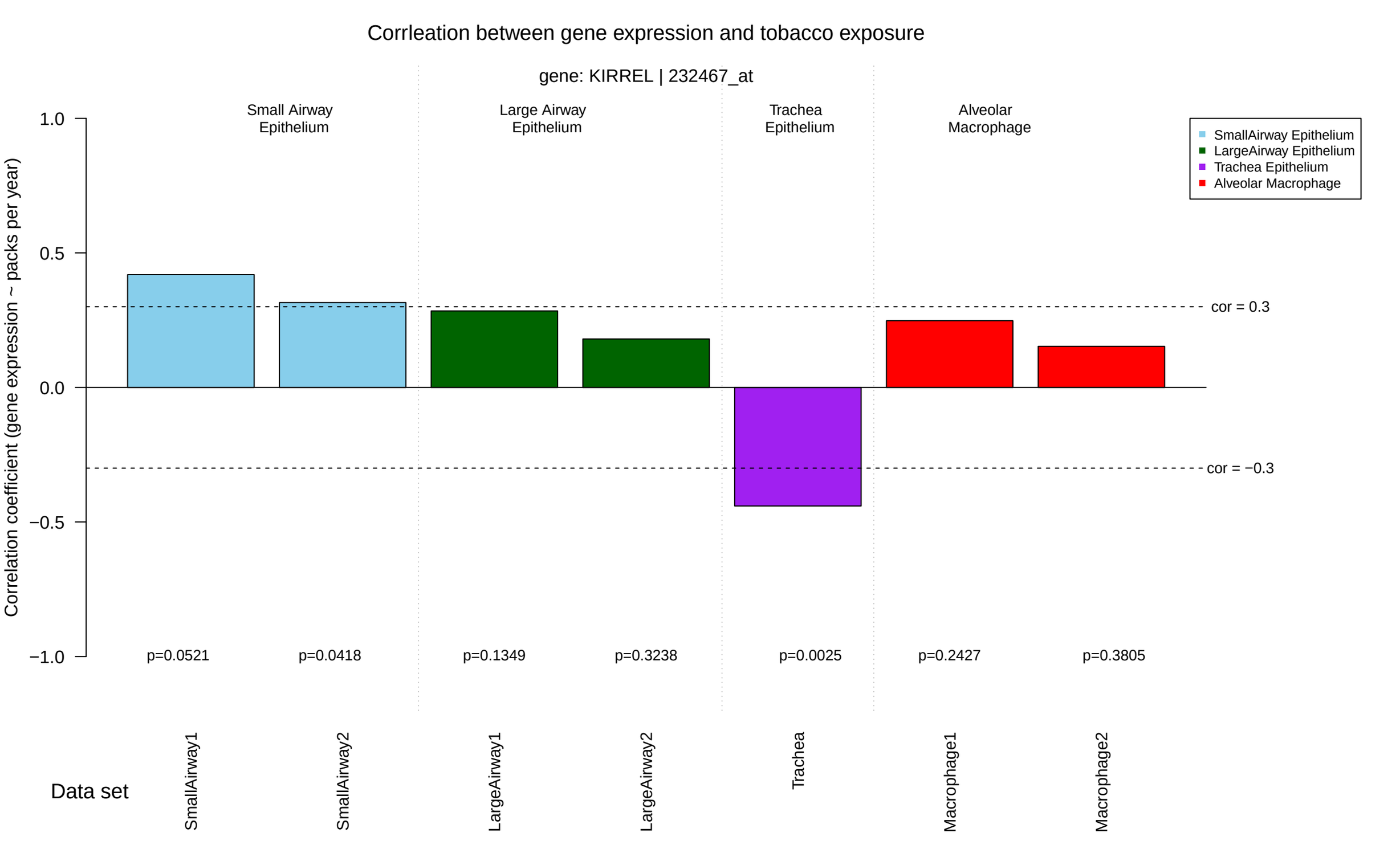

set: LargeAirway1 (p=0.0047)
KIRREL | 232467_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1667)
KIRREL | 232467_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0109)
KIRREL | 232467_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.02)
KIRREL | 232467_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1325)
KIRREL | 232467_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KIRREL | 232467_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0521
p=0.0418
p=0.1349
p=0.3238
p=0.0025
p=0.2427
p=0.3805
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2